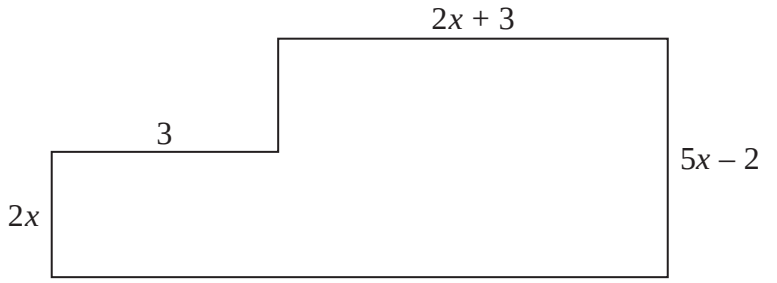
16 The diagram shows a shape.



All the corners are right angles.
All the measurements are given in centimetres.

The area of the shape is $14\,\text{cm}^2$

Find the value of x.

Area = $(2x + 3)(5x - 2) + 3(2x)$

1 mark
Split into the tall rectangle and lower-left rectangle.

$(2x + 3)(5x - 2) + 6x = 14$

1 mark
Add both areas and equate to 14.

$10x^2 - 4x + 15x - 6 + 6x = 14$

$10x^2 + 17x - 20 = 0$

1 mark
Expand and collect all terms on one side.

$(5x - 4)(2x + 5) = 0$

1 mark
Factorise the quadratic.

$x = 4/5$ or $x = -5/2$

A length 2x must be positive, so $x = 4/5$.

1 mark
Reject the negative solution: it gives negative lengths.

$x =$ 0.8

(Total for Question 16 is 5 marks)

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